

Task 6

Task Title:

Build a Context-Aware Chat Assistant with Memory

Task Description:

In this task, you will build a simple AI chat assistant that can remember information from previous messages and use it in future responses.

The goal is to understand how memory and context handling work in real Generative AI systems.

You will load a language model and create a conversation system that stores the conversation history and sends it back to the model as context for each new question.

This will allow the AI to maintain a conversation instead of responding to each message independently.

Task Objectives:

- Load an LLM using a Python library (such as Hugging Face Transformers)
- Build a chat loop where the user can ask multiple questions
- Store conversation history
- Inject the previous conversation as context in each new prompt
- Observe how memory affects the model's responses

Required Features

- Load a language model
- Maintain Conversation Memory (User messages and AI responses)
- Build Context for Each Prompt (Combine previous conversation history, Add the new user question, Send everything as context to the model)
- Run a Continuous Chat Loop (user: "My name is Ahmed.", ask:"What is my name?")